

1. The "Greasy Wrench" shop specializing in auto repairs rents a building at the corner of 77th and Moonawalker. In the short run, it can vary its output (the number of cars fixed) only by varying the number of mechanics it employs. Each Greasy Wrench employee is paid a fixed wage. Over the years, Wu Wei, the owner of "Greasy Wrench", has tried hiring different numbers of workers.

L	AP	TP	MP
1	5	5	5
2	7	14	9
3	10	30	16
4	11	44	14
5	10	50	6
6	9	54	4

- a. Complete the table by calculating the Total Product and Marginal Product corresponding to each number of labours.

Ans a. See the table. $TP = (AP) \times (\text{Number of labours})$; MP is the difference between two successive numbers in the TP column.

- b. At which worker does the Law of Diminishing Marginal Returns [DMR] start working? How do you know?

Ans b. Law of DMR starts working when MP starts decreasing. As we can see from the table, it happens when the firm hires the fourth worker (an alternative answer - "after the firm hires the third worker"; phrasing it either way is fine).

2. The information about the costs of a firm is given below.

Output	AFC, \$	AVC, \$
1	50.00	100.00
2	25.00	80.00
3	16.67	66.67
4	12.50	65.00
5	10.00	68.00
6	8.37	73.33
7	7.14	80.00
8	6.25	87.50

a. What is the firm's fixed cost?

Ans a. $FC = AFC \times Q$, and it doesn't matter which row of the table we take data from.
For instance, $\$50 \times 1 = \50 , and $\$6.25 \times 8 = \50 .

b. If the firm produces five units, what is the average total cost?

Ans b. $ATC = AFC + AVC$. For 5 units of output, $ATC = \$10 + \$68 = \$78$.

c. What is the total cost of producing four units?

Ans c. $TC = ATC \times Q = (AFC + AVC) \times Q$. For 4 units of output, $TC = (\$12.50 + \$65) \times 4 = \$310$.

Another way would be to calculate $VC = \$65 \times 4 = \260 and add it to $FC = \$50$ obtained in part a to get $TC = \$310$.

d. If the firm closes down and produces no output, what will be its total cost?

Ans d. No output means no variable cost. The firm's cost will then be limited to its fixed cost, $TC = FC = \$50$.